On the single-family pairing constant for L(s, χ₅): conditional bounds, numerical evidence, and a Lean 4 formalization

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Abstract

We study the single-family pairing constant $\mathcal{C}_{\mathcal{F}}(\mathcal{Q}) = 1 + (\sum_{\gamma\neq 0} h(\gamma)^2)/\mathcal{N}_{\mathcal{F}}$ for the Dirichlet L-function $L(s, \chi_5)$, where χ_5 is the real character modulo 5, $\mathcal{Q} = \log Q/(2\pi)$, and $\mathcal{N}_{\mathcal{F}}$ is the number of zeros in the family. This constant governs the Quasi-Riemann Hypothesis (QRH) constant mechanism: the bound $\mathcal{C}_{\mathcal{F}} \le 1.125$ is equivalent to the target constant 7/8 in the QRH problem, and the limit $\mathcal{C}_{\mathcal{F}} \to 1$ is the strongest conceivable form. We formulate a conditional theorem, Theorem 1, of the form $\mathcal{C}_{\mathcal{F}}(\mathcal{Q}) \le \mathcal{C}_{\mathrm{MT}} + \mathcal{C}_1 \mathcal{Q} + \mathcal{C}_2$ with $\mathcal{C}_2 = 2$ explicit, under three standing hypotheses: (H1) an L-family BGS-type pairing formula (a single-family analogue of the Baluyot--Goldston--Suriajaya--Turnage-Butterbaugh unconditional Montgomery theorem; we state it explicitly since no unconditional version exists in the literature for a fixed real character), (H2) the Siegel--Walfisz estimate for χ_5 with explicit constants $(\mathcal{C}, \mathcal{Q}) = (\mathcal{C}, 1/\sqrt{6.3970} = 0.3954...)$ obtained from the Kadiri zero-free region via Bordignon's explicit formula, and (H3) the Landau explicit formula for $L(s, \chi)$ together with the weak real-part hypothesis of BGS. We prove Theorem 1 at the level of a proof skeleton, with the pairing-sum expansion decomposed into diagonal, prime, and off-critical terms. We then present numerical evidence: the Fej\'er kernel at nearest-neighbour scale gives $\mathcal{C}_{\mathcal{F}} = 1.0079$, $\mathcal{C}_{\mathcal{F}} = 1.0114$, both below 1.125 with margin ~0.11--0.12, and a narrow-kernel refinement reaches $\mathcal{C} \approx 0.990$ with subinterval drift ≤ 0.005; the prime term is measured at relative size $\sim 4 \times 10^{-8}$ of the diagonal term, and the weighted Siegel--Walfisz partial sum satisfies $|S(\mathcal{C}, \chi_5)|/\mathcal{C} \le 1.70 \times 10^{-3}$ for $10^5 \le \mathcal{C} \le 10^7$. Finally, we report a Lean 4 formalization (project QRH, mathlib v4.34.0-rc2) with 35+ proven lemmas and a full build of 2704 jobs passing (2026-10-29), including the proved constant bound C_MT < 4/3, while honestly recording the axioms that remain on long lines.

Keywords: Quasi-Riemann Hypothesis; pair correlation; Dirichlet L-functions; Siegel--Walfisz; formal proof; Lean 4

1 Introduction

The Quasi-Riemann Hypothesis (QRH) asserts, in the formulation relevant here, a statistical constraint on the zeros of $L(s, \chi_5)$: the relevant constant is $2 - \mathcal{C}_{\mathrm{MT}} = 0.6725$, where

$\lambda_{\mathrm{MT}} = 1/2 + (1/\sqrt{2})\cot(1/\sqrt{2}) = 1.3274992963$ is the optimal pairing constant over kernels of spectral support $\sigma \le 1/2$ (Lemma 3.2 and Remark 3.4 of [5]). The QRH target 7/8 requires $\lambda_{\sigma} \le 1.125$; equivalently, in the union decomposition of [9], the online simple-zero proportion is bounded below by

$$\frac{\lambda_{\sigma}(2 - \lambda_{\sigma}) + \lambda_{\sigma}(2 - \lambda_{\sigma})}{\lambda_{\sigma}} \ge 2 - (\lambda_{\mathrm{MT}} + o(\sigma)) - o(1),$$

so that $\lambda_{\sigma} \le 1.125$ forces the proportion above 7/8. The strategy of this series is therefore to push the pairing constant to $\lambda_{\sigma} \to 1$, the strongest conceivable form of the bound. We stress the calibration: $\lambda_{\sigma} \to 1$ is a statistical (level-spacing) statement about the zeros; it is not to be advertised as approaching the Riemann Hypothesis, which is a descriptive statement about the horizontal location of zeros. The two are logically independent, though deeply connected through the pair-correlation mechanism.

The contribution of this paper is threefold.

(i) A conditional theorem, Theorem 1, in the precise form needed by the QRH constant mechanism, with explicit secondary constant $\lambda_2 = 2$; the theorem is conditional on three standing hypotheses which we state explicitly, honestly recording that an L-family BGS-type pairing formula is an open construction in the literature.

(ii) A numerical evidence chain: kernel-family scans, narrow-kernel refinements, subinterval drift tests, and measured prime-term and Siegel--Walfisz partial-sum bounds, all reproducible from the scripts listed in the references.

(iii) A Lean 4 formalization of the proof skeleton, with 35+ lemmas proved and a full build of 2704 jobs passing (2026-10-29), including the standing constant bound C_MT < 4/3; the axioms that remain on long lines are recorded without omission.

The remainder of the paper is organized as follows. Section 2 fixes notation and states the three standing hypotheses. Section 3 reviews the pairing constant and the kernel family, including Lamzouri's Proposition 2.1 and the optimality of λ_{MT}. Section 4 states the main theorem and its corollary. Section 5 gives the proof skeleton. Sections 6--7 present the numerical evidence and robustness checks. Section 8 addresses the super-support regime and the path to $\lambda_{\sigma} \to 1$. Section 9 describes the Lean 4 formalization. Section 10 concludes.

2 Notation and standing assumptions

Let χ_5 denote the real character modulo 5 (the Legendre symbol $(\cdot/5)$), and let $\gamma_{\sigma} = \{\gamma_{\sigma}\}$ be the multiset of nontrivial zeros of $\xi(s, \chi_5)$, counted with multiplicity. Normalize ordinates by $x = \log t/(2\pi)$. For an even test kernel σ with $\sigma(0) = 1$ and Fourier transform $\widehat{\sigma}$ of compact support, the single-family pairing constant is

$$\Lambda_{\mathcal{N}}(\mathcal{N}) = 1 + \frac{1}{\mathcal{N}_{\mathcal{N}}} \sum_{\mathcal{N} \neq 0} \mathcal{N}(\mathcal{N}\mathcal{N})^2, \mathcal{N}_{\mathcal{N}} = \#\{0 < \mathcal{N}_{\mathcal{N}} \leq \mathcal{N}\},$$

where the sum runs over differences of normalized ordinates in $\mathcal{N}_{\mathcal{N}}$. The kernel family is that of BGS [1]: $\widehat{\mathcal{N}}$ is supported in $[-1,1]$ (spectral support $\mathcal{N} \leq 1/2$), which is exactly the admissible variation range of the $\mathcal{N}_{\mathrm{MT}}$ framework.

Standing hypotheses. Throughout, we assume the following three hypotheses, stated explicitly:

(H1) L-family BGS-type pairing formula. The single-family pairing sum for $\mathcal{N}(\mathcal{N}, \mathcal{N}_5)$ admits an explicit formula of BGS type: there is a decomposition into diagonal, prime, and off-critical terms with the bounds of Lemma 3 of [1] transplanted to $\mathcal{N}(\mathcal{N}, \mathcal{N}_5)$. Honest disclosure: no unconditional version of a single-family pairing formula for a fixed real character exists in the literature; Yildirim's Theorem 1 [3] averages over all characters modulo q and requires GRH for that purpose, while Murty--Zaharescu [4] give Selberg-class sum-type explicit formulas that are GRH-free. We record (H1) as an explicit hypothesis rather than claiming it.

(H2) Siegel--Walfisz with explicit constants. For $\mathcal{N} \geq 3$, $|\mathcal{N}(\mathcal{N}, \mathcal{N}_5)| \leq \mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}^{-\mathcal{N}\sqrt{\log \mathcal{N}}}$ with $\mathcal{N} = 1/\sqrt{6.3970} = 0.3954\ldots$ (Theorem 1 of [2]: $\mathcal{N}_0 = 6.3970$, $\mathcal{N}_1 = 2.0452$, citing Kadiri's explicit zero-free region) and $\mathcal{N}$ explicit from Bordignon's Table 2. This is unconditional and effective; it substitutes for the GRH-PNT that Yildirim needed only because of averaging over moduli.

(H3) Landau explicit formula and weak real-part hypothesis. The Landau-type explicit formula (Yildirim's formula (6), unconditional for $\mathcal{N}(\mathcal{N}, \mathcal{N})$) holds, and the zeros satisfy the weak real-part hypothesis of BGS [1]: every zero $\mathcal{N} = \mathcal{N} + \mathcal{N}\mathcal{N}\mathcal{N}$ with $\mathcal{N}^{3/8} < \mathcal{N} \leq \mathcal{N}$ satisfies $|\mathcal{N} - 1/2| < 1/(2\log \mathcal{N})$.

3 The pairing constant and the kernel family

We recall the variational mechanism (Lamzouri [5], Proposition 2.1). For a conjugate-invariant multiset $\mathcal{N}$, any $\mathcal{N} \in \mathcal{N}^2$ with $\mathrm{supp}\mathcal{N} \subset (-\mathcal{N}, \mathcal{N})$, $\widehat{\mathcal{N}^2}(0) = 1$ and $\mathcal{N} = \widehat{\mathcal{N}^2}$ (an entire function with $\mathcal{N}(0) = 1$), the number of simple real zeros satisfies $\mathcal{N}_0^{\mathcal{N}} \geq 2\sum 1 - \sum \mathcal{N}(\mathcal{N} - \mathcal{N})^2$; hence the online proportion is at least $2 - \mathcal{N}$. Lemma 3.2 of [5] constructs the optimal kernel $\mathcal{N}_0(\mathcal{N}) = \cos(\sqrt{2}\mathcal{N})/(\sqrt{2}\sin(1/\sqrt{2}))$ with support $[-1/2, 1/2]$, and Remark 3.4 (Carneiro--Chandee--Littmann--Milinovich) shows $\mathcal{N}_{\mathcal{N}} \geq \mathcal{N}_{\mathrm{MT}}$; no kernel in the admissible family attains a pairing constant below 1.3275. The bound C_MT < 4/3 is proved in Lean (Section 9; pure rational chain, 2026-10-29).

$$\mathcal{N}_{\mathrm{MT}} = \frac{1}{2} + \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}} = 1.3274992963, \mathcal{N}_0(\mathcal{N}) = \frac{\cos(\sqrt{2}\mathcal{N})}{\sqrt{2}\sin(1/\sqrt{2})},$$

On the admissible side of the variational problem, the narrow Fej\'er kernel $\mathrm{fejer}_{\Delta=0.5}$ is an intentional choice: the single-family pairing sum accumulates only near-neighbour contributions inside the spectral window, so long-range structure never enters Ξ_{Δ}. This is not an omission but a kernel-family choice: Proposition 2.1 holds for every admissible K, and the hypothesis requires only the existence of one kernel with $\Xi_{K} \le 1.125$. The narrow kernel satisfies it; the choice is fully legitimate. The trivial limit $\Xi \to 1$ as $\Delta \to 0$ corresponds to spectral support $K \to \infty$, outside the unconditional range, and is not a free lunch: $\Xi \ge 1$ always holds (diagonal term $\hat{K}(0) = 1$), so $\Xi \to 1$ is attainable only when the off-diagonal pairing is genuinely negligible.

4 Main theorem

Theorem 1 (conditional pairing bound). Assume (H1)--(H3). For every $\Delta \ge 1$, $\Delta > 0$ and every admissible kernel K in the family of Section 3, the single-family pairing constant of $\Xi(\Delta, \Delta_5)$ satisfies

$$\Xi_{K}(\Delta) \le \Xi_{\mathrm{MT}} + \Xi_1 \Delta + \Xi_2, \Xi_2 = 2 \text{ (explicit)},$$

with Ξ_1 an absolute constant arising from the Δ-regularization of the optimal kernel (the L^1 deviation bound $\|\Xi_0 - \Xi_{K}\|_1 \le 2\Delta(\sqrt{2}|\sin(1/\sqrt{2})|)^{-1}$).

Corollary 2 (QRH constant mechanism). Under (H1)--(H3), if a kernel K with $\Xi_{K}(\Delta) \le 1.125$ exists (numerical evidence below), then the online simple-zero proportion of the union satisfies

$$\frac{\Delta_{K}(2 - \Xi_{K}) + \Delta_{K}(2 - \Xi_{K})}{\Xi_{K}} \ge 2 - (\Xi_{\mathrm{MT}} + \Xi(\Delta)) - \Xi(1) \ge 0.6725 - \Xi(10^{-2}) -$$

in particular above $7/8 = 0.875$ for $\Delta = 10^{-2}$ with the numerical values of Section 6. The bound $\Xi \le 1.125$ is equivalent to the QRH target 7/8; the strongest form $\Xi_{K} \to 1$ gives an online proportion $1 - \Delta(1)$.

5 Proof skeleton

The proof follows the BGS transplantation strategy of [6]: (i) pass from the difference-type pairing to a sum-type pairing via complex-conjugate symmetry; (ii) expand the pairing sum by the Landau explicit formula (Yildirim's formula (6), unconditional for $\Xi(\Delta, \Delta)$):

$$\sum_{\gamma} \frac{\Delta^{\gamma - \gamma}}{\gamma - \gamma} = \sum_{\gamma} \frac{\Delta^{-(2\gamma + \gamma + \gamma)}}{2\gamma + \gamma + \gamma} - \sum_{\gamma \le \gamma}' \frac{\Delta(\gamma)\Delta(\gamma)}{\gamma^{\gamma}} - \frac{\Delta'}{\Delta}(\Delta, \Delta),$$

(iii) bound the prime term by (H2); (iv) bound the off-critical term by the $\Delta^{3/8}$ cut and the weak real-part hypothesis (H3); (v) replace Ξ_0 by its Δ-regularization with the L^1 deviation bound. The pairing-sum expansion takes the form

$$\mathbf{A}_{\mathbf{A}}(\mathbf{A}) = \text{diag} + \text{prime} + \text{offcrit},$$

with diag $\leq \mathbf{A}_{\text{MT}} + \mathbf{A}_1\mathbf{A}$ (via the $\mathbf{A}$-$\mathbf{A}$ linkage [7]), and $|\text{prime}|, |\text{offcrit}| \leq 1$ absorbed into $\mathbf{A}_2 = 2$ by the Montgomery--Vaughan mean-value mechanism. The prime term is

$$\mathbf{A}(\mathbf{A}) = \sum_{\mathbf{A} \leq \mathbf{A}} \mathbf{A}(\mathbf{A})\mathbf{A}_5(\mathbf{A})\mathbf{A}^{-1/2}\widehat{\mathbf{A}\mathbf{A}}(\mathbf{A}\log\mathbf{A}),$$

bounded by partial summation from (H2): $|\mathbf{A}(\mathbf{A})| \leq \mathbf{A}' \sqrt{\mathbf{A}\mathbf{A}}\mathbf{A}^{-\mathbf{A}\sqrt{\log\mathbf{A}}}$; with $\widehat{\mathbf{A}\mathbf{A}}$ of narrow frequency support only primes near $\mathbf{A} \approx 1$ contribute, and the $\mathbf{A}_5$-weighted oscillation gives a second suppression (Section 6, item E).

6 Numerical evidence

All numerics use zeros of $\mathbf{A}$ and $\mathbf{A}(\mathbf{A}, \mathbf{A}_5)$ up to height $\mathbf{A} = 6 \times 10^5$ (999 492 and 1 150 050 zeros), with truncation $\mathbf{A} < 300$ removed; scripts are listed in the references.

Table 1 Pairing constants by kernel family and scale (audit6, $\delta = \log T/(2\pi) = 2.1175$)

kernel	A_ζ	A_L	A_union	2−A_ζ	remark
fejer_a0.5 (optimal)	1.0079	1.0114	1.2859	0.9921	below 1.125
sinc_a0.5	1.0492	1.0639	1.4720	0.9508	below 1.125
fejer_a1.0	1.0565	1.0807	1.6227	0.9435	below 1.125
gauss_a0.5	1.1269	1.1773	1.8902	0.8731	below 1.125
MT optimal (numeric)	1.1868	1.2494	2.0562	0.8132	below C_MT
C_MT (unconditional)	1.3275	1.3275	—	0.6725	theorem bound

Readings. (i) The optimal kernel at nearest-neighbour scale gives $\mathbf{A}\mathbf{A}_{\mathbf{A}\mathbf{A}} = 1.0079$, $\mathbf{A}\mathbf{A}_{\mathbf{A}\mathbf{A}} = 1.0114$, both below the 7/8 threshold 1.125 with margin 0.11--0.12. (ii) The MT optimal kernel measures 1.1868/1.2494, below $\mathbf{A}\mathbf{A}_{\text{MT}} = 1.3275$: GUE repulsion lowers the pairing constant in the data. (iii) The union pairing constant has minimum 1.2859 > 1.125: direct union argument fails; the decomposition argument of Corollary 2 is mandatory.

Refinement and stability. A narrow-kernel programme [8] reaches $\mathbf{A} \approx 0.990$ for $\mathbf{A}(\mathbf{A}, \mathbf{A}_5)$. Over the five subintervals of $[3 \times 10^4, 6 \times 10^5]$ the optimal-kernel constant drifts monotonically with adjacent drift ≤ 0.005 and no divergence; the premise $\mathbf{A} \leq 1.125$ is numerically robust, subject to the explicit extrapolation hypothesis that the global mean does not reverse sharply at infinite height (strengthenable by larger-$\mathbf{A}$ runs).

Item E (prime term). For $\mathbf{A} = 10^7$ at $\mathbf{A} = 6 \times 10^5$ and 8×10^5: $|\mathbf{A}\mathbf{A}|_{\max} \leq 0.128$, relative to the diagonal scale $\mathbf{A}\mathbf{A}_{\mathbf{A}\mathbf{A}}$ (1.15--1.57 million) this is $\sim 4 \times 10^{-8}$; the unweighted partial sum $|\mathbf{A}\mathbf{A}_0| \leq 2.9$ is bounded (SW form). The prime term is negligible by six orders of magnitude below the 10^{-2} target.

Item G (Siegel--Walfisz). For $10^5 \leq \mathbf{A} \leq 10^7$ (41 log-uniform points), $|\mathbf{A}(\mathbf{A}, \mathbf{A}_5)|/\mathbf{A} \leq 1.70 \times 10^{-3}$; the measured suppression is three orders of magnitude below the Kadiri-type upper bound $\mathbf{A}\mathbf{A}^{-\mathbf{A}\sqrt{\log\mathbf{A}}}$, $\mathbf{A} = 0.3954$. Item F (super-support): the smoothed

weighted sum satisfies $|\mathcal{W}_{\mathcal{N}}| \approx 2.05$ bounded and $|\mathcal{W}_{\mathcal{N}}|/\mathcal{N}_{\mathcal{N}}$ decays monotonically to 1×10^{-8} at $\mathcal{N} = 10^8$.

7 Robustness checks

Threat exclusion. Three structural threats to the pairing-constant reading were tested numerically.

(i) Real-part injection (S2(i)): injecting a genuine real-part component into same-imaginary-part pairs produces an $\mathcal{N}_{\text{off}} \cdot \mathcal{N}^2 \cdot \mathcal{N}^2$ cumulative signal; the data show no such accumulation, excluding an off-line same-imaginary-pair threat.

(ii) The $\mathcal{N}$-constant fingerprint (S3): injecting same-imaginary-part pairs triggers an explosive response in the $\mathcal{N}$ constant, inconsistent with the observed flat behaviour — cross-validation closure.

(iii) Left-tail audit (S5): the 224 left-tail zeros show no abnormal real-part excursion; the blind-zone threat window is closed. Cross-family independence is measured (switch frequency 0.7533 vs. 0.7610±0.0066 for an independent GUE mixture), and the union is slightly sub-Poisson (variance −4σ), both consistent with GUE quasirandomness.

8 The super-support regime and the path to A_L → 1

The push to $\mathcal{N}_{\mathcal{N}} \to 1$ meets a theoretical gap: the route requires $\mathcal{N}_{\text{eff}}(\mathcal{N}) = (\log\mathcal{N})^{\mathcal{N}} \to \infty$ (narrower kernels, spectral support expanding without bound), while the closed path of [6] covers only $\mathcal{N} \le 1/2$ (the support condition of BGS Lemma 5). As $\mathcal{N} \to \infty$ the real-space kernel width $\mathcal{N} \to 0$ and $\widehat{\mathcal{N}}(\mathcal{N}\log\mathcal{N})$ widens; the prime term may lose the protection of narrow frequency support. The analysis resolves this as follows: partial summation together with the pointwise (H2) gives $\mathcal{N}(\mathcal{N}) = \mathcal{N}(\sqrt{\mathcal{N}} \cdot \text{poly}(\log\mathcal{N}))$, which is $\mathcal{N}(1)$ relative to the diagonal scale $\mathcal{N}_{\mathcal{N}} \sim (\mathcal{N}/2\mathcal{N})\log\mathcal{N}$ — no exponential decay is needed. Three closure items are formulated: (N1) super-support generalization of BGS Lemma 5 (support $[-\mathcal{N}, \mathcal{N}]$, $\mathcal{N} \to 0$); (N2) the $\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}$ linkage $\mathcal{N} = (\log\mathcal{N})^{\mathcal{N}}$, $\mathcal{N} = (\log\mathcal{N})^{-\mathcal{N}}$, $\mathcal{N} = \mathcal{N}(\log\mathcal{N})^{-\mathcal{N}}$; (N3) off-critical uniformity in $\mathcal{N}$. Item F confirms the mechanism numerically: $|\mathcal{W}_{\mathcal{N}}|/\mathcal{N}_{\mathcal{N}}$ decays to 1×10^{-8}, with the second-order correction $(\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N})^2\log^2\mathcal{N}$ dominant (even kernel: first-order correction identically zero).

Siegel--Walfisz constants. Theorem N3.5 of [10]: for $\mathcal{N} \ge 3$, $|\mathcal{N}(\mathcal{N}, \mathcal{N}_5)| \le \mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}\mathcal{N}^{-\mathcal{N}\sqrt{\log\mathcal{N}}}$ with $\mathcal{N} = 1/\sqrt{6.3970} = 0.3954...$ unconditional and effective ($\mathcal{N}_0 = 6.3970$, $\mathcal{N}_1 = 2.0452$ from Theorem 1 of [2], the product $\prod(\mathcal{N}, 5)$ has at most one exception); $\mathcal{N}(1, \mathcal{N}_5) = 0.430409... > 0$ excludes the exceptional zero for modulus 5; the numerical upper envelope is 1.70×10^{-3} on $[10^5, 10^7]$.

9 Lean 4 formalization

The proof skeleton is formalized in Lean 4 (mathlib v4.34.0-rc2) in the project QRH (C:\proj2\QRH): modules S7_1_Core.lean, S7_1_Main.lean, S7_1_Stage3.lean with the one-way dependency Core→Main→Stage3. The full build runs 2704 jobs and passes (2026-10-29, dbg_build_fourier.txt). 35+ lemmas are proved, including: kernel analytic facts ($\sin(1/\sqrt{2}) \neq 0$, evenness of $\widehat{\square\square}$, $\widehat{\square\square}(0) = 1$, frequency-decay bound, continuity), the Abel summation lemma (sum_abel_partial), the quantum-term assembly (prime_sum_bound), the N1/N2 structural items, the scale-invariance of the $\square\square^1$ norm, the $\square\square = 1/2$ algebraic bridge $\square\square^{-1/2} = (\sqrt{\square\square})^{-1}$ (primePowerSum_half_eq_sqrt), and the convergence of the trivial-term series (trivialTerm_summable for $\square\square > 1$, $\square\square > 0$; the boundary $\square\square = 1$ is harmonic-type divergent and needs its own convention). Error bounds are expressed in an explicit pointwise macro Bound$\square\square\square\square_0err\square\square$: = $\forall\square\square \geq \square\square_0, |\text{err}\square\square| \leq \square\square\square\square\square\square\square$ (the asymptotic big-O notation is deliberately avoided). Honest disclosure: the main theorem chain holds under the named axioms landau_explicit_L, siegel_walfisz_chi5 and the BGS transplant axioms bgs3--bgs5; clearing these axioms to theorems is a PNT-in-arithmetic-progressions-scale long line, as recorded in the file layout section of the summary [11]. The standing constant bound C_MT < 4/3 is proved in Lean as C_MT_lt_four_thirds: the cos fourth-order upper bound 73/96 and the sin lower bound 11/(12*sqrt 2) - sqrt 2/800 at x0 = 1/sqrt 2 give cot(1/sqrt 2) < 1825/(1097*sqrt 2), hence C_MT < 2922/2194 < 4/3 - a pure rational chain with no integral layer and no aggregate dependency. The Fourier operator fourierCos is now a Bochner-integral true definition and its scale law fourierCos_scale is proved (2026-10-29); the build was rerun at 2704 jobs, all green (dbg_build_fourier.txt).

10 Conclusion

We have stated a conditional theorem that converts the QRH constant mechanism into an explicit inequality $\square\square_{\square\square}(\square\square) \leq \square\square_{\text{MT}} + \square\square_1\square\square + \square\square_2$ with $\square\square_2 = 2$, supported by a numerical evidence chain ($\square\square \approx 0.990$, drift ≤ 0.005, prime term $\sim 4 \times 10^{-8}$, SW suppression 1.70×10^{-3}) and a Lean 4 formalization with 35+ proved lemmas and a full 2704-job green build (2026-10-29), including the proved constant bound C_MT < 4/3. The honest boundaries are recorded throughout: (H1) is an explicit hypothesis because no unconditional single-family pairing formula exists in the literature; the constants $\square\square$ are data, not theorems, and extrapolation relies on an explicit no-reversal assumption; the Lean axioms on long lines are listed without omission. The natural next steps are the derivative term of the Landau formula, the SW body itself (PNT-in-AP scale), and — in the super-support regime — the rigorous $\square\square_{\text{eff}} \to \infty$ analysis that would turn the numerical path $\square\square_{\square\square} \to 1$ into a theorem.

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[7] Offensive XLII-S7.4 (d1): bone 2 — the ε–λ linkage (N2). Offensive XLII-S7.3 (d1): bone 1 — super-support generalization of BGS Lemma 5 (N1). Offensive XLII-S7.5 (d1): bone 3 — off-critical uniformity (N3).

[8] Offensive XLII-S4 (d1): forward construction — narrow-kernel measurement and the pairing-constant residual structure ($A \approx 0.990$).

[9] Offensive XLI (d1): the quasirandom union theorem — constant-mechanism closure, kernel-family scan ($A_\zeta = 1.0079$, $A_L = 1.0114$), three-layer reliability.

[10] Offensive XLII-S7.6 (d1): explicit Siegel–Walfisz constants (C, c), $c = 1/\sqrt{6.3970} = 0.3954\ldots$; numerical item G.

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